NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Spine

1. cervical trauma
2. thoracolumbar trauma
3. degenerative cervical myelopathy
4. lumbar disc and stenosis

Cervical trauma

1. Clearance and imaging

- **NEXUS criteria** (no imaging needed if all five absent): **midline cervical tenderness, focal neurological deficit, altered alertness, intoxication, painful distracting injury**.
- **Canadian C-spine rule:** high-risk factors (**age 65 or more, dangerous mechanism, paraesthesia in the limbs**) = image; low-risk factors allowing range-of-motion assessment; able to rotate the neck 45 degrees left and right = no imaging. More sensitive and specific than NEXUS.
- **CT is the first-line imaging** in adult trauma (radiographs miss injuries, especially at the cervicothoracic junction and craniocervical junction). **MRI** for neurological deficit, suspected ligament/disc injury (**before reducing a facet dislocation in an obtunded patient**), SCIWORA, and for obtunded patients (debated: CT alone is sufficient to clear many obtunded patients with a high-quality negative CT).
- **Normal values: atlantodens interval (ADI) up to 3 mm in adults (5 mm in children)**; prevertebral soft tissue **under 6-7 mm at C2** and **under 22 mm at C6**; **Powers ratio** (BC/OA) under 1 (over 1 = anterior atlanto-occipital dislocation); **basion-dens interval and basion-axial interval under 12 mm (Harris "rule of 12")**.

2. Craniocervical junction

- **Occipital condyle fractures (Anderson-Montesano):** I comminuted (impaction), II extension of a basilar skull fracture, **III avulsion by the alar ligament** (potentially unstable). Most: collar; occipitocervical fusion if unstable.
- **Atlanto-occipital dissociation:** usually fatal; **children** more often (large head, horizontal condyles). **Traynelis types** (anterior,

longitudinal/distraction, posterior). Detect with **Powers ratio, BDI/BAI**, CT; **never apply traction**; occipitocervical fusion.

3. Atlas (C1) fractures

- **Jefferson (burst) fracture: axial load**, four-part ring fracture (anterior and posterior arches). **Rule of Spence:** combined lateral mass overhang **over 6.9 mm (about 7 mm; 8.1 mm with radiographic magnification)** on the open-mouth view suggests **transverse atlantal ligament (TAL) rupture** (unstable). MRI shows the TAL directly; **Dickman classification of TAL injury** (type I midsubstance rupture: needs **C1-C2 fusion**; type II bony avulsion of the TAL insertion: may heal with immobilisation).
- **Treatment:** stable (intact TAL): **rigid collar/halo**; unstable: **halo** or **C1-C2 (or occipitocervical) fusion**.
- **Neurological injury is rare** (canal widens).

4. Axis (C2) fractures

Odontoid (dens) fractures: Anderson-D'Alonzo (with Grauer modification)

Type	Location	Stability / union	Treatment
I	**Tip avulsion (alar ligament)**, above the transverse ligament	Usually stable (exclude **atlanto-occipital dissociation**)	Collar
II	**Junction of dens and body (waist)**	**Commonest; highest non-union** (watershed blood supply, small contact area)	See below
III	Extends into the **C2 body** (cancellous)	Good union	**Collar or halo**

- **Grauer subtypes of type II: IIA** transverse undisplaced (collar/halo), **IIB** oblique from **anterosuperior to posteroinferior** (or transverse

displaced) = suitable for **anterior odontoid screw** (fracture line perpendicular to the screw), **IIC** oblique from **anteroinferior to posterosuperior** (or comminuted) = **posterior C1-C2 fusion** (screw would shear the fracture).

- **Non-union risk factors for type II: displacement over 5 mm** (also over 4-6 mm), **posterior displacement**, **angulation over 10 degrees**, **age over 50 (about 40-60 in studies)**, smoking, delay in treatment, **comminution**.
- **Elderly (most common cervical fracture over 65):** halo poorly tolerated (aspiration, pneumonia, mortality); options: **rigid collar** (accept stable fibrous union), or **posterior C1-C2 fusion** (Harms: **C1 lateral mass + C2 pars/pedicle screws**; Magerl **C1-C2 transarticular screws**; Gallie/Brooks wiring) for younger elderly patients fit for surgery.
- **Anterior odontoid screw:** needs an **intact TAL**, reducible fracture with a favourable obliquity, and a body habitus allowing the trajectory (**short neck, barrel chest, kyphosis contraindicate it**); osteoporotic bone fails. Preserves C1-C2 rotation (C1-C2 provides about **50% of cervical rotation**).
- **Os odontoideum:** smooth, well-corticated ossicle separated from a hypoplastic dens (probably old unrecognised fracture or developmental); unstable if **sagittal plane translation**: posterior C1-C2 fusion if instability or neurological signs.

Hangman's fracture (traumatic spondylolisthesis of the axis)

- **Bilateral pars (isthmus) fractures of C2**; mechanism **hyperextension with axial load** (diving, car crashes; judicial hanging is hyperextension-distraction). **Neurological injury is uncommon** (canal widens).
- **Levine-Edwards classification:**

Type	Features	Mechanism	Treatment
I	**Under 3 mm translation, no angulation**	Hyperextension + axial load	**Rigid collar**
II	**Over 3 mm translation + significant angulation** (C2-3 disc disrupted)	Hyperextension then flexion	**Reduction (traction in extension) + halo**
IIA	**Severe angulation, minimal translation** (oblique fracture; C2-3 disc and PLL disrupted)	**Flexion-distraction**	**Avoid traction (worsens it)**; reduce with **compression and extension**, halo
III	Pars fracture + **bilateral C2-3 facet dislocation**	Flexion-compression	**Open reduction and fusion** (C2-3 posterior or anterior)

5. Subaxial cervical spine (C3-C7)

Classification

- **SLIC (Subaxial Injury Classification and Severity score, Vaccaro):**

Component	Score
Morphology: no abnormality 0; **compression 1**; burst +1 (=2); **distraction 3**; **rotation/translation 4**	0-4
Disco-ligamentous complex: intact 0; **indeterminate 1**; **disrupted 2**	0-2
Neurology: intact 0; **root injury 1**; **complete cord injury 2**; **incomplete cord injury 3**; **continuous cord compression in the setting of a neurological deficit +1**	0-4

  - **Total under 4: non-operative; 4 = surgeon's discretion; over 4 (5 or more): surgery.**

- **AO Spine subaxial classification:** A (compression), B (tension band injury), C (translation), with facet injury and neurology modifiers.
- **Allen-Ferguson (mechanistic):** compressive flexion, vertical compression, distractive flexion (facet dislocations), compressive extension, distractive extension, lateral flexion.

Specific injuries

- **Facet dislocations (distractive flexion):**
  - **Unilateral facet dislocation:** about **25% anterolisthesis**; rotation; **radiculopathy** commonly; often stable-ish but needs reduction.
  - **Bilateral facet dislocation:** about **50% or more anterolisthesis**; **high rate of cord injury**; very unstable.
  - **Reduction: awake, cooperative patient: closed traction reduction** (Gardner-Wells tongs, incremental weights about 5 lb per level up to about 10 lb/level plus 10 lb for the head, serial neurological examination and radiographs) **is safe without pre-reduction MRI**. **Obtunded/uncooperative patient or failed closed reduction: MRI first** to exclude a **disc herniation** behind the vertebral body (could be pushed into the cord), then open reduction (anterior discectomy first if a disc is present). Then **stabilisation** (anterior cervical discectomy and fusion, or posterior lateral mass fixation).
- **Burst fractures (vertical compression):** retropulsion; anterior corpectomy and fusion if cord compression.
- **Teardrop fracture (flexion-compression): anteroinferior vertebral body fragment** with posterior displacement of the remaining body into the canal and **posterior ligament disruption**: **very unstable**, often **anterior cord syndrome**. Distinguish from the **extension teardrop** (small anteroinferior avulsion of C2 by the ALL in extension: usually stable).

- **Clay-shoveler's fracture:** spinous process avulsion (C7, C6, T1); stable.
- **Ankylosing spondylitis / DISH: extension injuries through the fused spine** ("**chalk-stick**" fractures) are **highly unstable**, frequently missed, often with **epidural haematoma** and late neurological deterioration. **CT/MRI the whole spine; never force into extension** (immobilise in the pre-injury kyphotic position); usually **long posterior fixation** (three levels above and below).
- **Central cord syndrome** (hyperextension in an elderly spondylotic spine, no fracture): see below.
- **Vertebral artery injury:** with **foramen transversarium fractures** and facet dislocations: **CT angiography**; antiplatelet/anticoagulant therapy if no contraindication.

6. Spinal cord injury (SCI)

Assessment

- **ASIA (ISNCSCI) examination: 10 key muscles** (C5 elbow flexors, C6 wrist extensors, C7 elbow extensors, C8 finger flexors (middle finger DIP), T1 little finger abductors; L2 hip flexors, L3 knee extensors, L4 ankle dorsiflexors, L5 long toe extensors, S1 ankle plantarflexors) and **28 sensory dermatomes** (light touch, pin prick), plus **deep anal pressure and voluntary anal contraction (sacral sparing)**.
- **ASIA Impairment Scale:**

Grade	Definition
A	**Complete**: no motor or sensory function in S4-S5
B	Sensory incomplete: sensory but no motor function below the level, including S4-S5
C	Motor incomplete: motor function below the level, **more than half** of key muscles below the level **grade under 3**

Grade	Definition
D	Motor incomplete: **at least half** of key muscles below the level **grade 3 or more**
E	Normal

- **Spinal shock:** temporary loss of all cord function and reflexes below the injury (flaccid paralysis, areflexia), **ends with the return of the bulbocavernosus reflex** (usually within **24-48 hours**). **Completeness can only be judged after spinal shock resolves**: no sacral sparing after the bulbocavernosus reflex returns = complete injury (very poor prognosis for recovery).
- **Neurogenic shock:** loss of sympathetic tone (injuries above **T6**): **hypotension with bradycardia**, warm peripheries. Treat with fluids then **vasopressors** (noradrenaline/phenylephrine), atropine for bradycardia. **Haemorrhagic shock** shows tachycardia and must be excluded first.
- **Maintain mean arterial pressure 85-90 mmHg for about 7 days** (AANS/CNS guideline).
- **Methylprednisolone: not routinely recommended** (NASCIS II/III results not reproducible; complications: infection, GI bleeding). AO Spine 2017 guideline: may be considered within 8 hours (24-hour infusion) as an option; never after 8 hours.
- **Early decompression (within 24 hours)** improves neurological recovery (**STASCIS** study, cervical SCI; AO Spine guideline recommends surgery within 24 hours).

Incomplete cord syndromes

Syndrome	Mechanism	Features	Prognosis
Central cord	**Hyperextension in an elderly spondylotic spine** (often no fracture)	**Upper limbs weaker than lower** (hands worst), sacral sparing, variable sensory loss	**Good**; bowel/bladder and walking usually recover (hands least); **commonest incomplete syndrome**
Anterior cord	Flexion, anterior compression, **anterior spinal artery** injury	**Motor loss and loss of pain/temperature**; **preserved proprioception and vibration (dorsal columns)**	**Worst** prognosis of the incomplete syndromes
Brown-Sequard	Penetrating injury, hemisection	**Ipsilateral motor loss and proprioception loss; contralateral pain and temperature loss** (one to two levels below)	**Best prognosis** (about 90% walk)
Posterior cord	Rare (hyperextension)	Loss of proprioception and vibration only	Good
Conus medullaris	T12-L1 injury	Bowel/bladder/sexual dysfunction, saddle anaesthesia, mixed UMN/LMN; absent bulbocavernosus reflex may persist	Variable
Cauda equina	Below L1	**LMN** pattern: asymmetric lower limb weakness, areflexia, saddle anaesthesia, sphincter dysfunction	Better than conus (peripheral nerves)

Functional levels (complete injuries)

- **C3 and above:** ventilator-dependent (phrenic C3-5).
- **C5:** elbow flexion; power wheelchair (hand control), feeds with adapted devices.
- **C6: wrist extension (tenodesis grip)**; manual wheelchair with adaptations, independent transfers possible.
- **C7: elbow extension (triceps)**: independent transfers and manual wheelchair; **key level for independent living**.
- **C8-T1:** hand function; full wheelchair independence.
- **Upper limb tendon transfers** in tetraplegia: **posterior deltoid or biceps to triceps** (elbow extension), **brachioradialis to ECRB** (wrist extension), **brachioradialis to FPL** (key pinch), nerve transfers.
- **Autonomic dysreflexia** (injuries above T6): noxious stimulus below the injury (**full bladder, faecal impaction**, pressure sores) causes **severe hypertension with bradycardia, headache, flushing/sweating above the lesion**: sit the patient up, remove the stimulus (catheterise), antihypertensives.

7. Paediatric cervical spine

- **Under about 8 years: C1-C3 injuries predominate** (large head, horizontal facets, ligamentous laxity, fulcrum at C2-3); **pseudosubluxation of C2 on C3 (up to about 4 mm)** is normal (**Swischuk line**: posterior arch line C1 to C3 should pass within 2 mm of the C2 spinolaminar line); **ADI up to 5 mm**; **SCIWORA**.
- Use a **backboard with an occipital cut-out or shoulder roll** to avoid neck flexion.
- **Atlantoaxial rotatory subluxation (Fielding-Hawkins types I-IV)**; **Grisel syndrome** (after pharyngeal infection or surgery); **cock-robin** torticollis; dynamic CT. Treat early with collar/traction, halo or fusion for chronic cases.

Exam pearl: **Rule of Spence (lateral mass overhang over about 7 mm) = TAL rupture; Dickman I (midsubstance) needs fusion. Odontoid type II = commonest and highest non-union; IIB = anterior screw, IIC = posterior C1-C2 fusion. Hangman IIA: no traction.**

Exam pearl: **Awake patient with facet dislocation: closed reduction without MRI is safe; obtunded or failed: MRI first. SLIC over 4 = surgery.** Ankylosing spondylitis fractures are highly unstable: CT/MRI whole spine.

Exam pearl: **Central cord (elderly hyperextension, arms worse than legs) is commonest; Brown-Sequard best prognosis; anterior cord worst.** Spinal shock ends with return of the **bulbocavernosus reflex**; neurogenic shock = hypotension + bradycardia (above T6). **C7 = key level for independence.**

Thoracolumbar trauma

1. Anatomy and biomechanics

- The **thoracolumbar junction (T11-L2)** is the commonest site of spinal fractures: transition from the rigid, rib-supported, kyphotic thoracic spine (coronal facets) to the mobile, lordotic lumbar spine (sagittal facets).
- The **thoracic canal is narrow** (and the cord blood supply tenuous: watershed **T4-T9**; **artery of Adamkiewicz** usually left-sided **T9-L2**), so thoracic injuries cause more **complete cord injuries**. The **cord ends (conus) at L1-L2**; below this, injuries affect the **cauda equina** (better prognosis).
- **Denis three-column concept: anterior** (ALL, anterior two-thirds of the body and disc), **middle** (posterior third of body and disc, **PLL**), **posterior** (pedicles, facets, laminae, **posterior ligamentous complex: supraspinous, interspinous ligaments, ligamentum flavum, facet capsules**). **Instability when two or more columns fail** (middle column key in Denis's model).
- The **posterior ligamentous complex (PLC)** is the posterior **tension band**; its integrity is the main determinant of stability in modern classifications. MRI: **STIR hyperintensity** in interspinous region, disrupted black line of the supraspinous ligament/ligamentum flavum; radiographs/CT: **widened interspinous distance**, facet diastasis/perch, **splaying of spinous processes**.

2. Classifications

Denis (1983)

- **Compression** (anterior column only; stable), **burst** (anterior + middle; types A-E), **flexion-distraction (seat-belt/Chance)**, **fracture-dislocation** (all three columns).

TLICS (Thoracolumbar Injury Classification and Severity score, Vaccaro)

Component	Points
Morphology: compression 1, **burst 2**, **translation/rotation 3**, **distraction 4**	1-4
Neurology: intact 0; **nerve root 2**; **complete cord/conus 2**; **incomplete cord/conus 3**; **cauda equina 3**	0-3
PLC: intact 0; **suspected/indeterminate 2**; **injured 3**	0-3

- **Under 4: non-operative; 4: either; over 4 (5 or more): operative.**

AO Spine thoracolumbar classification (2013)

- **Type A (compression):** A0 minor/process fractures; **A1 wedge** (one endplate); **A2 split/pincer** (both endplates, coronal split); **A3 incomplete burst** (one endplate + posterior wall); **A4 complete burst** (both endplates + posterior wall).
- **Type B (tension band failure): B1 transosseous posterior tension band disruption (bony Chance)**; **B2 posterior ligamentous tension band disruption** (with any A injury of the body); **B3 hyperextension** (anterior tension band disruption: ankylosed spines).
- **Type C (displacement/translation in any direction):** all elements fail; **the most unstable**.
- **Neurology modifiers:** N0 intact, N1 transient, N2 radicular, N3 incomplete cord/cauda, N4 complete cord, NX unexaminable; **case modifiers M1 (indeterminate PLC injury), M2 (patient-specific comorbidity: ankylosing spondylitis, osteoporosis, burns)**.
- **TL AOSIS** score guides treatment (similar to TLICS).

3. Specific injuries and their management

Compression (wedge) fractures

- Anterior column only; **stable**. **Non-operative** (brace or early mobilisation as pain allows). Concern if **loss of anterior height over 50%**, **kyphosis over 20-30 degrees**, or multiple contiguous fractures (consider PLC injury).
- **Osteoporotic vertebral compression fractures:** commonest osteoporotic fracture; often asymptomatic; **MRI STIR** shows acute oedema; treat pain, bone health (anti-osteoporotic therapy; **teriparatide** speeds healing). **Vertebroplasty/kyphoplasty** for persistent severe pain (evidence mixed: **sham-controlled trials (Buchbinder, Kallmes) showed no benefit over sham**; some trials (VERTOS IV/ II) differ). **Kümmell disease:** delayed post-traumatic vertebral collapse with an **intravertebral vacuum cleft** (AVN).

Burst fractures

- **Axial load**; anterior + middle column failure with **retropulsion** of the posterior wall into the canal; **interpedicular widening** on AP radiograph.
- **Neurologically intact burst fracture with intact PLC (TLICS 2-3, AO A3/A4 N0): non-operative treatment** (early mobilisation with or without a TLSO brace) gives **similar outcomes to surgery** (**Wood et al.**, RCT: no difference at long-term follow-up; **brace vs no brace (Bailey 2014)**: no difference). **Canal compromise percentage and kyphosis alone do not dictate surgery**; retropulsed fragments remodel (resorption of canal fragments).
- **Surgical indications: neurological deficit with canal compromise**, **PLC disruption**, progressive kyphosis/neurology, **kyphosis over 25-30 degrees** or **height loss over 50%** (relative, debated), polytrauma requiring mobilisation.
- **Surgery: posterior pedicle screw fixation** (short-segment: one level above and below ± screws in the fractured vertebra (**intermediate screws**) improve kyphosis correction; long segment for unstable/osteoporotic), **indirect decompression by ligamentotaxis** (works only if the **PLL is intact** and within about 4 days), **direct**

decompression (posterolateral transpedicular or anterior corpectomy) for significant canal compromise with deficit; **anterior corpectomy and reconstruction** for severe comminution (**load-sharing classification, McCormack: score 7 or more** = anterior column reconstruction needed; points for comminution, apposition, kyphosis correction).

Flexion-distraction (Chance) injuries

- **Lap belt** (axis of flexion anterior to the spine): **tension failure of all three columns**, horizontal fracture through spinous process, lamina, pedicles and body (**bony Chance**, B1) or through ligaments/disc (**ligamentous**, B2).
- **Associated intra-abdominal injuries in about 40-50%** (bowel perforation, mesenteric tears): **"seat-belt sign"** on the abdomen: CT abdomen, surgical review.
- **Treatment: purely bony Chance fracture** can heal in a **hyperextension cast/brace** (if neurologically intact and reduced); **ligamentous or disc injuries do not heal** reliably: **posterior compression instrumentation and fusion**.

Fracture-dislocations (AO C)

- Shear, rotation or translation; **highest rate of neurological injury** (complete cord injury common); **posterior instrumentation (long-segment) with decompression and fusion** in all.

Hyperextension injuries in ankylosed spines (AS, DISH)

- **Long-lever fractures through the fused spine**, highly unstable, frequently missed (**image the whole spine with CT ± MRI**), **epidural haematoma** with delayed neurological deterioration. **Immobilise in the pre-injury (kyphotic) position**; **long posterior fixation (at least 2-3 levels above and below)**. High mortality in the elderly.

4. Sacrum and lumbosacral junction (overlap with pelvis)

- **Spinopelvic dissociation (U/H-type sacral fractures):** falls from height/suicidal jumpers ("**jumper's fracture**"); bilateral vertical sacral fractures with a transverse component; cauda equina injury; **lumbopelvic (triangular) fixation** ± decompression.
- **Roy-Camille** classification of transverse sacral fractures.

5. Neurological injury management (see cervical notes)

- **ASIA examination**; spinal shock (bulbocavernosus reflex); maintain **MAP 85-90 mmHg** for about 7 days; **early decompression (within 24 hours)** where feasible; **no routine steroids**.
- **Conus medullaris syndrome** (T12-L1): mixed UMN and LMN signs, early bowel/bladder involvement, saddle anaesthesia; **cauda equina syndrome** (below L1-L2): LMN pattern, better recovery.
- Rehabilitation: bladder (intermittent catheterisation), bowel programmes, pressure care, DVT prophylaxis (high VTE risk: LMWH within 72 hours), heterotopic ossification (hips), autonomic dysreflexia (above T6), spasticity, psychological care.

6. Penetrating spinal injury

- **Gunshot wounds:** steroids are **contraindicated** (more complications); **surgery rarely improves neurology**; remove bullets only for **progressive deficit**, **CSF leak**, **bullet within the canal at T12-L5 (cauda equina: possible benefit)**, **lead toxicity** risk if in the disc/CSF, or infection; **transcolonic injuries need broad-spectrum antibiotics** (debridement debated).

7. Paediatric thoracolumbar injuries

- **Chance fractures with lap belts** (with abdominal injuries), **apophyseal ring fractures** (slipped vertebral apophysis: adolescent; behaves like a disc herniation; may need excision), **SCIWORA**

(commoner in younger children), **non-accidental injury** (multiple compression fractures). Post-traumatic **scoliosis/kyphosis** after SCI in young children (almost universal before puberty).

8. Complications

- **Post-traumatic kyphosis** (painful; corrective osteotomy: **pedicle subtraction osteotomy** about 30-40 degrees, Smith-Petersen about 10 degrees per level), **non-union/implant failure** (short-segment fixation without anterior support in comminuted fractures), **post-traumatic syrinx** (late neurological deterioration: MRI), **infection**, **dural tears**, **adjacent segment degeneration**, VTE, pressure sores.

Exam pearl: **TLICS: morphology (compression 1, burst 2, translation 3, distraction 4) + neurology + PLC; over 4 = surgery. Neurologically intact burst fracture with intact PLC = non-operative (Wood RCT), with or without a brace.**

Exam pearl: **Chance = lap belt flexion-distraction; 40-50% have abdominal visceral injury; bony Chance can heal in extension cast, ligamentous needs fusion. Ligamentotaxis needs an intact PLL. Load-sharing score 7 or more = anterior reconstruction.**

Exam pearl: **AO B = tension band failure (B1 bony Chance, B2 ligamentous, B3 hyperextension); AO C = translation (most unstable).** Ankylosed spine fractures: whole-spine imaging, long posterior fixation.

Degenerative cervical myelopathy

1. Definitions and pathology

- **Degenerative cervical myelopathy (DCM)** is the umbrella term for cord compression from degenerative disease: **cervical spondylotic myelopathy (CSM)** (disc-osteophyte complexes, ligamentum flavum buckling/hypertrophy, facet hypertrophy, degenerative spondylolisthesis), **OPLL** (ossification of the posterior longitudinal ligament), **ossification of the ligamentum flavum**, and congenital stenosis. **Commonest cause of spinal cord dysfunction in adults over 55.**
- **Static factors:** congenitally narrow canal (**Torg-Pavlov ratio under 0.8**: canal diameter/vertebral body width; **absolute AP canal diameter under 13 mm** = relative stenosis, **under 10 mm** = absolute), disc-osteophyte bars, OPLL.
- **Dynamic factors: extension** narrows the canal (ligamentum flavum buckles; pincer between posteroinferior body and the lamina below); **flexion** stretches the cord over anterior bars; listhesis on flexion-extension films.
- **Ischaemic/cellular injury:** anterior spinal artery territory compromise, demyelination, apoptosis of oligodendrocytes; grey matter (anterior horn) and corticospinal tract involvement.
- **Commonest levels: C5-C6, then C6-C7** (most mobile).

2. Clinical features

- **Symptoms: clumsy hands** (difficulty with buttons, writing, dropping objects), **numb hands** (non-dermatomal), **gait imbalance** (broad-based, spastic, worse in the dark), neck stiffness/pain (not always), Lhermitte sign (electric shock with neck flexion), late **bladder dysfunction**.

- **Upper motor neurone signs:** hyperreflexia, **Hoffmann sign** (flicking the middle finger DIP produces thumb/index flexion), **inverted brachioradialis (radial) reflex** (tapping the brachioradialis gives finger flexion and reduced brachioradialis contraction: lesion at C5-6), **Babinski**, clonus, spasticity, **scapulohumeral reflex** (C3-4 or above), **pectoralis reflex**.
- **"Myelopathy hand" (Ono): finger escape sign** (little and ring fingers drift into abduction/flexion when the fingers are held extended and adducted; ulnar intrinsic weakness), **grip-and-release test** (normal over 20 times in 10 seconds), intrinsic wasting.
- **Gait:** tandem walking difficulty, **Romberg** positive (dorsal column).
- **Radiculopathy may coexist (myeloradiculopathy).**
- **Differentials: motor neurone disease (ALS)** (no sensory signs, fasciculations including the tongue, EMG), **multiple sclerosis** (brain lesions, optic neuritis, oligoclonal bands), **subacute combined degeneration (B12)**, syringomyelia, **normal pressure hydrocephalus** (gait, incontinence, dementia), tumours, peripheral neuropathy, **carpal tunnel syndrome** (may coexist), hereditary spastic paraparesis, copper deficiency, HIV myelopathy, tropical spastic paraparesis (HTLV-1).

3. Severity scoring

- **Modified Japanese Orthopaedic Association (mJOA) score** (18 points: upper limb motor 5, lower limb motor 7, sensory 3, bladder 3): **mild 15-17, moderate 12-14, severe under 12**.
- **Nurick grade (0-5)** based on gait and employment. Others: Cooper, 30-metre walk test, grip and release, Neck Disability Index.

4. Imaging

- **Radiographs:** spondylosis, alignment (**lordosis vs kyphosis**: affects choice of approach), **Torg-Pavlov ratio**, flexion-extension instability, **OPLL** (visible on lateral).

- **MRI (investigation of choice):** cord compression and **intramedullary signal change**:
  - **T2 hyperintensity alone** may be reversible (oedema).
  - **T1 hypointensity with T2 hyperintensity** indicates **myelomalacia/cavitation** (irreversible): **worse prognosis**.
  - "**Snake-eyes**" (bilateral anterior horn T2 hyperintensity on axial images): poor prognosis.
  - Compression ratio, cross-sectional area of the cord.
- **CT: OPLL** extent and type, **double-layer sign** (suggests **dural ossification**: dural tear/CSF leak risk with anterior removal), facet arthropathy, bone quality; **CT myelography** where MRI is contraindicated.
- **OPLL types: continuous, segmental (commonest), mixed, circumscribed (focal)**. Common in **East Asians** (Japan about 2-4%), associated with **DISH**, diabetes, obesity. **Occupying ratio over 60%** and **K-line negative** (OPLL crosses the K-line: line connecting midpoints of the canal at C2 and C7 on lateral) predict **failure of posterior decompression**: use an anterior approach or posterior fusion.
- **Neurophysiology** (SSEPs, MEPs) in uncertain cases and to exclude ALS/neuropathy.

5. Natural history and treatment decisions

- DCM is **usually progressive in a stepwise fashion**; about 20-60% deteriorate without surgery. **Surgery halts progression and often improves function**; outcome depends on **pre-operative severity and duration of symptoms**.
- **AO Spine / WFNS guideline (Fehlings 2017):**
  - **Moderate and severe DCM (mJOA under 15): surgery recommended.**

  - **Mild DCM (mJOA 15-17): surgery or a supervised trial of structured rehabilitation**; operate if it worsens or fails to improve.
  - **Non-myelopathic patients with cord compression on MRI** but radiculopathy (clinical or electrophysiological): higher risk of developing myelopathy: counsel; surgery or close follow-up.
  - **Asymptomatic cord compression without radiculopathy: no prophylactic surgery**; counsel and follow.
- **Predictors of poor outcome: long duration of symptoms**, **severe pre-operative mJOA**, **older age**, **T1 hypointensity/myelomalacia on MRI**, smoking, **multilevel** disease, gait impairment.

6. Surgical options

Anterior approaches (Smith-Robinson; between the carotid sheath laterally and the trachea/oesophagus medially)

- **ACDF (anterior cervical discectomy and fusion):** for **1-2 level disease** with disc-level compression; restores lordosis; direct removal of disc-osteophyte bars.
- **Anterior corpectomy and fusion (ACCF):** compression behind the vertebral body (retrovertebral), focal OPLL, kyphotic multilevel disease. **Hybrid ACDF-ACCF** constructs.
- **Indications for anterior surgery: kyphotic alignment**, **1-3 level** compression, **anterior pathology** (disc, focal OPLL), **K-line negative OPLL**.
- **Complications: dysphagia** (commonest, usually transient; more with multilevel surgery), **recurrent laryngeal nerve palsy** (hoarseness; classically **right-sided** approach (longer variable course of the right RLN) though evidence is mixed), **oesophageal perforation** (rare, serious), **vertebral artery injury** (lateral dissection beyond the uncovertebral joints), **C5 palsy**, **pseudarthrosis** (higher with **multilevel ACDF** and corpectomies),

graft dislodgement (**multilevel corpectomy: add posterior fixation**), **adjacent segment degeneration** (about 2-3% per year, Hilibrand), airway compromise from haematoma (**emergency evacuation** at the bedside), CSF leak (OPLL).

Posterior approaches

- **Laminoplasty** (open-door (**Hirabayashi**), French-door (**Kurokawa**)): **multilevel (3 or more) disease** with **preserved lordosis**; preserves motion; **contraindicated in kyphosis** (cord cannot drift back) and in instability. **Complications: axial neck pain**, **loss of range of motion**, **C5 palsy** (about 5-10%).
- **Laminectomy and fusion (lateral mass screws ± C7/T1 pedicle screws):** multilevel disease with **lordosis or neutral alignment**, instability, axial neck pain; **laminectomy alone is avoided** (**post-laminectomy kyphosis**, especially in young patients and in children).
- **Posterior surgery relies on dorsal drift of the cord**, which needs **lordotic (or at least neutral) alignment** (**cervical lordosis over about 10 degrees**, or **K-line positive** in OPLL).

Approach choice summary

Situation	Preferred approach
1-2 levels, anterior compression	**Anterior (ACDF)**
Kyphosis, focal retrovertebral compression	**Anterior (corpectomy)** or combined
3 or more levels, lordotic, no neck pain	**Laminoplasty**
3 or more levels, lordotic/neutral, instability or neck pain	**Laminectomy + fusion**
OPLL K-line negative / occupying ratio over 60%	**Anterior** or posterior fusion
Multilevel with kyphosis	**Combined anterior-posterior**

C5 palsy

- **Deltoid (± biceps) weakness** appearing **hours to days after surgery** (anterior or posterior), about **5%** (higher after laminoplasty/laminectomy-fusion and corpectomy); proposed mechanisms: **tethering of the C5 root with dorsal cord drift**, foraminal stenosis, reperfusion injury. **Most recover within 6-12 months**; prophylactic **C4-5 foraminotomy** debated.

7. Cervical radiculopathy (related)

- Commonest **C6 (C5-C6 disc)** and **C7 (C6-C7 disc)**. Root = number of the lower vertebra (C5-C6 disc compresses C6 root).

Root	Motor	Reflex	Sensory
C5	Deltoid, biceps	**Biceps**	Lateral arm
C6	**Wrist extensors**, biceps	**Brachioradialis**	Thumb, index, radial forearm
C7	**Triceps**, wrist flexors, finger extensors	**Triceps**	**Middle finger**
C8	Finger flexors	None	Little finger, ulnar forearm
T1	Hand intrinsics	None	Medial arm

- **Spurling test** (extension + rotation + axial compression reproduces radicular pain), **shoulder abduction relief sign** (Bakody).
- **Natural history is favourable**: about 75-90% improve without surgery (analgesia, physiotherapy, selective nerve root injections). **Surgery** for progressive/significant motor deficit or intractable pain after about 6-12 weeks: **ACDF**, **cervical disc arthroplasty** (motion preserving; 1-2 levels without facet arthropathy, instability or OPLL; similar or better outcomes and lower adjacent-level reoperation in

trials), or **posterior foraminotomy** (foraminal soft disc or osteophyte, no instability).

Exam pearl: **Hoffmann sign, inverted radial reflex, finger escape sign, broad-based gait = myelopathy. T1 hypointensity + T2 hyperintensity = myelomalacia (poor prognosis). Moderate/severe DCM (mJOA under 15) = surgery.**

Exam pearl: **Laminoplasty needs lordosis (contraindicated in kyphosis); K-line negative OPLL = anterior. C5 palsy: deltoid weakness after decompression, usually recovers. Dysphagia is the commonest complication of anterior cervical surgery.**

Lumbar disc and stenosis

1. Disc degeneration

- **Intervertebral disc: nucleus pulposus** (type II collagen, high **proteoglycan (aggrecan)** and water content; notochordal origin) and **annulus fibrosus** (type I collagen outer layers, lamellae at about 30 degrees alternating). **Largest avascular structure in the body**: nourished by diffusion through the **vertebral endplates**.
- **Degeneration:** loss of proteoglycans and water, **increased type I : type II collagen ratio**, fissures, **MMPs and cytokines (IL-1, TNF-alpha)**; smoking, genetics (strongest factor in twin studies), age.
- **MRI grading: Pfirrmann (I-V)** for disc degeneration; **Modic endplate changes:**

Modic	T1	T2	Meaning
I	Low	**High**	**Oedema/inflammation**; most associated with **low back pain**; possible low-grade infection (Cutibacterium acnes)
II	**High**	Iso/high	**Fatty marrow replacement**; commonest; stable
III	Low	Low	**Sclerosis**

2. Lumbar disc herniation

Terminology and location

- **Bulge** (over 25% of circumference) vs **herniation**: **protrusion** (base wider than the dome), **extrusion** (dome wider than base, or beyond the disc level), **sequestration** (free fragment).
- **Location: central**, **paracentral (posterolateral, commonest about 90%)**, **foraminal (far lateral, about 5-10%)**, extraforaminal.
- **Root affected:** a **paracentral L4-L5 herniation compresses the traversing L5 root**; a **far lateral (foraminal) L4-L5 herniation**

compresses the exiting L4 root.

- **Commonest levels: L4-L5 and L5-S1 (about 95%)**; peak age 30-50.

Clinical

Root	Motor	Reflex	Sensory
L4	Quadriceps, tibialis anterior	**Knee jerk**	Medial leg/medial malleolus
L5	**EHL** (big toe extension), **gluteus medius** (hip abduction: Trendelenburg), tibialis posterior, ankle dorsiflexors	None reliable (medial hamstring)	**Dorsum of foot, first web space**
S1	Gastrocnemius-soleus (toe walking), peronei, gluteus maximus	**Ankle jerk**	Lateral foot, sole, little toe

- **Straight-leg raise (Lasègue)** positive between 30-70 degrees (L5/S1 roots); **crossed SLR** (pain in the affected leg when lifting the other) is **highly specific**. **Femoral stretch test** for L2-L4 (upper lumbar/foraminal herniations).
- **Red flags:** cauda equina features, progressive motor deficit, fever/weight loss/cancer history (infection/tumour), trauma, age over 50 with new pain, night pain.

Natural history and treatment

- **Favourable:** most improve within 6-12 weeks; **extruded and sequestrated fragments regress most** (resorption; inflammatory response).
- **Non-operative:** activity modification (avoid bed rest), analgesics/NSAIDs, neuropathic agents (limited evidence), physiotherapy, **transforaminal epidural steroid injection** (short-term relief).

- **SPORT trial (Spine Patient Outcomes Research Trial):** both groups improved; **surgery gave faster and greater improvement** in pain and function (as-treated analysis), with high crossover between groups (intention-to-treat inconclusive).
- **Surgical indications: cauda equina syndrome (emergency)**, **progressive or severe motor deficit** (e.g. foot drop: earlier surgery may improve recovery), **intractable radicular pain after about 6 weeks of conservative care** with concordant imaging.
- **Surgery: microdiscectomy** (standard; interlaminar approach), endoscopic discectomy; **far lateral herniation: Wiltse paraspinal (muscle-splitting between multifidus and longissimus) approach** to avoid facetectomy.
- **Complications: recurrent herniation (about 5-15%)**, **dural tear** (about 1-7%: repair primarily; headache, pseudomeningocele), wrong-level surgery, nerve root injury, **discitis**, rare **vascular injury (anterior annulus perforation: aorta/iliac vessels; hypotension: emergency laparotomy)**.

Cauda equina syndrome (CES)

- **Massive central disc (commonest cause, L4-L5/L5-S1)**, also tumour, haematoma, abscess, trauma.
- **Features: bilateral sciatica**, **saddle anaesthesia** (perianal/perineal), **bladder dysfunction (painless urinary retention with overflow is late: CES-R)**, bowel dysfunction, sexual dysfunction, reduced anal tone.
- **Classification: CES-S** (suspected: bilateral radiculopathy only), **CES-I** (incomplete: altered urinary sensation, reduced desire to void, poor stream, but no retention), **CES-R** (retention with overflow; worse prognosis).
- **Bladder scan: post-void residual over about 200 ml** raises concern (sensitive but not specific).

- **Emergency MRI and decompression as soon as possible** (outcomes better when operated **before retention (CES-I)**; **within 48 hours** classically (Ahn meta-analysis), with current UK/GIRFT guidance for **urgent surgery at the earliest opportunity**).

3. Lumbar spinal stenosis

- **Central stenosis** (ligamentum flavum hypertrophy/buckling, facet hypertrophy, disc bulge, listhesis), **lateral recess** (subarticular; superior articular process hypertrophy compresses the traversing root), **foraminal** (exiting root; disc height loss, osteophytes). **L4-L5 commonest.**
- **Neurogenic claudication:** leg pain/heaviness/paraesthesia with walking and **standing (extension)**, relieved by **sitting or bending forward (flexion)** (**shopping-cart sign**, cycling tolerated better than walking), proximal-to-distal progression, **normal pulses**.
- **Vascular claudication:** pain in the calf with walking, **relieved by standing still** (no need to sit), absent pulses, distal-to-proximal, **cycling also painful**, low ABI.
- **Imaging:** MRI (canal cross-sectional area **under about 75-100 mm²** = stenosis; **AP diameter under 10 mm** = absolute central stenosis; **sedimentation sign**, **Schizas grading** A-D).
- **Treatment:** physiotherapy (flexion-based), analgesia, epidural injections (limited benefit in central stenosis: **LESS trial**: epidural steroid not better than lidocaine alone). **Decompression (laminectomy / laminotomy, undercutting facetectomy preserving over 50% of facets)** for persistent disabling symptoms: **SPORT stenosis cohort** and **Malmivaara** trial showed **surgery better than non-operative care**.
- **Fusion with decompression:** for **degenerative spondylolisthesis with instability**, deformity, or iatrogenic instability. **Two 2016 NEJM trials: Försth (Swedish Spinal Stenosis Study)**: **adding fusion gave no benefit** in stenosis with or without stable spondylolisthesis;

Ghogawala (SLIP): fusion gave **slightly better outcomes and fewer reoperations** in **stable grade I degenerative spondylolisthesis**. Result: individualise; fuse when **dynamic instability (over 3-4 mm translation or over 10-15 degrees angulation on flexion-extension)**, high disc height, facet effusion, or sagittal imbalance.

- **Interspinous spacers (X-Stop):** higher reoperation rates than decompression; limited role.

4. Spondylolysis and spondylolisthesis

Wiltse-Newman classification

Type	Name	Features
I	**Dysplastic (congenital)**	Dysplastic L5-S1 facets, **elongated pars**; high risk of progression and **cauda equina** symptoms (intact posterior arch pulls on the roots)
II	**Isthmic**	**Pars defect**: IIA **lytic (stress fracture; commonest in children/adolescents)**, IIB elongated healed pars, IIC acute fracture. **L5-S1 commonest**; **L5 root** compressed in the foramen
III	**Degenerative**	**Women over 50, L4-L5 commonest**, intact pars, facet arthropathy with **sagittal facets**; **L5 root** (lateral recess) and central stenosis
IV	Traumatic	Acute fracture of the posterior elements other than the pars
V	Pathological	Tumour, infection, bone disease
VI	Post-surgical (iatrogenic)	After excessive decompression

- **Meyerding grade** (percentage slip of the upper vertebra on the lower): **I under 25%, II 25-50%, III 50-75%, IV 75-100%, V over 100% (spondyloptosis)**. **Low grade = I-II; high grade = III or more.**

- **Spinopelvic parameters** (see spinopelvic notes): **high pelvic incidence** predisposes to isthmic slip and progression; **slip angle** (kyphosis at L5-S1) over about 45-50 degrees predicts progression in children.
- **Spondylolysis in adolescents (athletes: gymnasts, cricket fast bowlers, footballers):**
  - Extension-related low back pain; **Scotty dog collar sign** on oblique radiographs; **SPECT/CT, or MRI (STIR oedema) for early stress reaction** (no radiation).
  - Treatment: **activity restriction (about 3 months)** ± brace; most become pain-free even if the defect does not unite. **Direct pars repair** (Buck screw, Scott wiring, pedicle screw-hook) for symptomatic L1-L4 defects/young patients without disc degeneration or slip.
- **Isthmic spondylolisthesis in children: progression is commoner in girls, before or during the growth spurt, with high-grade or dysplastic slips**. **Fusion (L4-S1 or L5-S1 in situ)** for **slip over 50%**, progression, or persistent symptoms. **Phalen-Dickson sign:** knee-flexed, hip-flexed gait with tight hamstrings.
- **Adult isthmic and degenerative spondylolisthesis:** decompression ± fusion (posterolateral, interbody: TLIF/PLIF/ALIF/LLIF). **Reduction of high-grade slips** risks **L5 nerve root palsy** (stretch).

5. Low back pain: key evidence and syndromes

- **Non-specific low back pain:** most resolve; **avoid imaging without red flags**; stay active; NICE: exercise, manual therapy alongside exercise, CBT-based approaches; **avoid opioids/paracetamol alone**; radiofrequency denervation for facet pain in selected patients.
- **Yellow flags:** psychosocial predictors of chronicity (fear-avoidance, depression, compensation).
- **Discogenic pain/fusion:** fusion vs intensive rehabilitation trials (**Fairbank/MRC**: no significant difference; Fritzell (Swedish) showed

benefit of fusion). **Disc arthroplasty** for single-level discogenic pain in select patients.

- **Facet joint pain**, **sacroiliac joint dysfunction** (diagnostic blocks), **piriformis syndrome**, **Baastrup disease** (kissing spines), **Bertolotti syndrome** (L5 transverse process pseudoarticulation with sacrum).
- **Failed back surgery syndrome:** wrong diagnosis, wrong level, recurrent herniation, epidural fibrosis, instability, infection, adjacent segment disease; **spinal cord stimulation** for neuropathic leg pain.

Exam pearl: **Paracentral L4-L5 disc = traversing L5 root; far lateral L4-L5 = exiting L4 root (Wiltse approach). L5 = EHL and first web space; S1 = ankle jerk, lateral foot. Crossed SLR is the most specific sign.**

Exam pearl: **Cauda equina: saddle anaesthesia, bilateral sciatica, urinary dysfunction: emergency MRI and decompression, ideally before retention (CES-I). SPORT: surgery gives faster and greater improvement for disc herniation and stenosis.**

Exam pearl: **Neurogenic claudication relieved by flexion/sitting (shopping cart, cycling ok); vascular relieved by standing still. Degenerative spondylolisthesis: women over 50, L4-L5, sagittal facets. Isthmic: L5-S1, pars defect, Scotty dog collar.** Meyerding I-V by 25% steps.